



EL MAHDI ATMANI

AI Engineering | Data Scientist | NLP & OSINT

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🌐 EL Mahdi ATMANI 🏠 Github 📁 Portfolio

Profile

Specialist in AI Engineering and Data Science with experience building end-to-end machine learning systems and scalable data pipelines. Proven expertise in fine-tuning Large Language Models and Vision-Language Models to drive intelligent automation and solve complex technical challenges.

Skills

Cloud & Computing

- **Cloud Platforms:** DigitalOcean, AWS, Wasabi, OpenStack
- **Parallel Computing:** OpenMP for algorithm optimization

Databases & Data Management

- **Databases:** SQL, Oracle, NoSQL, MongoDB, DynamoDB, neo4j, Snowflake

Programming

- **Programming Languages:** Python, C++, Java, Android Studio, XML
- **Web Development:** JSP, HTML, CSS, JavaScript, Flask, JEE

Artificial Intelligence & Data Science

- **Machine Learning:** Python (scikit-learn, NumPy, pandas, Matplotlib), Performance measurement
- **Deep Learning:** TensorFlow, PyTorch, Keras
- **Neural Networks:** CNN, RNN, NN
- **Computer Vision:** YOLO, OpenCV
- **Natural Language Processing:** RAG, fine-tuning, Hugging Face, spaCy, NLTK, or LangChain
- **Image & Signal Processing:** MATLAB
- **Data Visualization & Analytics:** Power BI, Matplotlib
- **Automation:** n8n, scripts
- **AI/LLMs:** LangChain/LangGraph, Multi-Agent Systems, RAG Architectures, LLMs (Claude, Gemini, OpenAI, Mistral ...)

Professional Experience

Apr/2025 – Nov/2025
Rabat, Morocco

AI/NLP Engineer Intern

Moroccan Parliament

Intelligent System for International Diplomatic Watch

- Built an end-to-end intelligent OSINT platform, automating diplomatic data collection across 12 international media sources and APIs for the Moroccan Parliament.
- Fine-tuned a Llama 3.1 (8B) model using PEFT/LoRA (13M parameters) for domain-specific NLP tasks, achieving 98.39% accuracy in relevance classification.
- Optimized local GPU deployment by implementing 4-bit quantization, reducing the model's memory footprint by 75% while maintaining high inference speeds.
- Implemented a full-stack solution with a Next.js/React frontend and a Python backend, integrating Playwright for robust web scraping and TinyDB to process and categorize diplomatic events into 52 thematic areas.
- **Technologies:** Llama 3.1, PEFT/LoRA, PyTorch, LangChain, Ollama, Next.js, Playwright, Python, TinyDB

Jan/2025 – Present
Rabat, Morocco

AI Training Specialist & NLP Evaluator

Outlier.ai : Freelance

LLM Fine-Tuning and Model Alignment

- Conducted rigorous evaluation and validation of Large Language Models (LLMs) to enhance instruction-following capabilities, logical reasoning, and factual accuracy.
- Applied Reinforcement Learning from Human Feedback (RLHF) methodologies by analyzing and grading AI outputs for coherence, context retention, and safety alignment.
- Designed and executed complex, multi-turn prompting strategies (including edge-case testing) to identify system vulnerabilities, mitigate hallucinations, and establish model boundaries.
- Curated, structured, and validated high-quality, domain-specific training datasets to support the continuous fine-tuning of foundational generative models.
- **Technologies:** LLMs, RLHF, Prompt Engineering, NLP, Model Alignment, Data Quality Assurance

Jun/2021 – Jul/2021
Beni Mellal, Morocco

Dev Intern

Regional Hospital

- Developed a comprehensive patient management application to digitize medical records, streamline admissions, and improve administrative workflow efficiency for healthcare professionals.

- Provided responsive IT support and technical problem-solving for hospital staff, ensuring the continuous and reliable operation of critical hardware and software systems.
- **Technologies:** Java, JavaFX, MySQL

Education

Feb/2024 – Dec/2025 Kenitra, Morocco	Master of Research in Computer Science and Artificial Intelligence. <i>Faculty of Science - Ibn Tofail</i> 
Feb/2021 – Sep/2021 Taroudant, Morocco	Professional Bachelor's Degree Computer Engineering. <i>Polydisciplinary Faculty - Ibn Zohr</i> 

Projects

Complex OCR & Document Extraction Developed an end-to-end Optical Character Recognition system utilizing a fine-tuned Vision-Language Model (Gemma-3 4B) capable of extracting structured text, tables, and detailed metadata from complex scanned legal documents.
Web Scraping AI Agent Pioneered an AI-driven web scraping agent that pinpointed news content across 750+ websites, liberating the team from manual research and saving 80% of the time spent on data collection.
Deutsch b'Darija AI Language Learning Platform Developed an educational platform utilizing Retrieval-Augmented Generation (RAG) to generate dynamic, context-aware German language instruction combined with gamified elements for Moroccan users.

Languages

• Arabic : Native	• French : Fluent	• English : Fluent
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